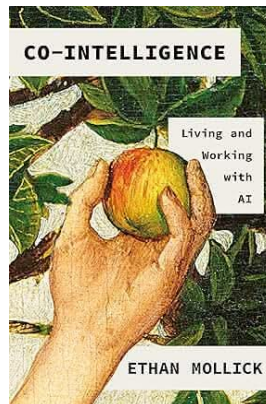


Co-Intelligence: Living and Working with AI

April 2024 by Ethan Mollick



Part I: The Book

Three Days of Sleepless Nights

- Experiencing ChatGPT gave him (or anybody) **three days of sleepless nights.**
 - 5 years of work (tens of people with tons of money) to build simulation systems
 - 80% of the work instantly done with prompts.
- Students begin to stop asking questions in class.
 - Perfect grammar.
 - Wrong (absurd) references

- We don't know the answer how to live and work with AI yet.
- This book is an attempt to figure it out and give practical advice (guidance) - "Invite & Control"

Navigating the Age of AI

- AI is becoming **increasingly important** in every field.
- Mark Zuckerberg predicted **2025 as a decisive year** for AI.
- The message: **embrace AI** as a critical skill, not something to avoid.

AI as a "Dual Brain (Co-intelligence)"

- Use AI as a **secondary cognitive resource** — a *dual brain*.
- Core principles:
 - i. **Integrate AI** into your workflows. (Let's call it Invite)
 - ii. Keep **human control** in decision-making. (Let's call it Control)

Practical Applications

- AI can **enhance productivity and creativity**.
- Example: ChatGPT simplifies complex concepts, helps with writing & coding.
- Case study:
 - A detailed business simulation built with ChatGPT.
 - Normally: requires a large team + weeks of work.
 - With AI: **faster, cheaper, and effective**.

- Some professionals are adapting AI to **improve their work** dramatically.
 - They focus on the practical use of AI rather than theoretical concerns.

AI Capabilities & Limits

- **Large Language Models (LLMs):** predict text using tokens.
- Strengths: generate ideas, explain concepts, accelerate tasks.
- Limits:
 - *Hallucinations* → AI produces false or nonsensical info.
 - Requires **human fact-checking & oversight.**

Ethics & Future of AI

- Concerns: bias, misuse, and safety.
- **AI Alignment:** ensuring AI benefits humanity.
- Future roles of AI:
 - Collaborator
 - Teacher
 - Coach
- Human creativity, judgment, and empathy remain **irreplaceable.**

Key Takeaways

- AI is not just a tool — it is a **partner in thinking**.
- Learn to **use AI effectively** while staying critical.
- Balance **potential and limitations** for personal & professional growth.

Part 2: Interpretation of this Book as a Software Engineer

The Two Rules (Invite & Control) from the Book

1. Always invite AI to **think with you**, not just for you.
 - **AI is a cognitive partner** — not just a simple tool give you answers.
2. Always **remain in control** of decisions.
 - i. Humans must **validate and oversee** AI outputs, it's always your responsibility in the end.

Why These Rules Matter

1. AI is not the answering machine; it is a **thinking partner**.

- It can help you generate ideas, but it cannot replace your judgment.

2. If we delegate all thinking to AI, our brain will wrongly think that it is no longer needed.

- This leads to **cognitive atrophy** — we lose our ability to think critically and creatively.

Problem: Cognitive Atrophy

It's a horrible situation if we become **over-reliant on AI** and lose our own cognitive abilities.

We should learn to work with AI as a "dual brain"

- Think of AI as a "**dual brain**" that enhances human thinking with your check and balance.
- Think of it as a **genius assistant (but sometimes insane)** who can help you brainstorm, write, and code, but who also makes mistakes and needs your supervision.

Solution: Building the System for Invite & Control

- AI is a cognitive partner: We need a **systematic way** to work with it effectively.
- Humans must remain in control: We need to build the system to **oversee and validate** AI outputs.

We have two choices:

1. Be the 10x, or 100 x super software engineer who can leverage AI to do things that were impossible before.
2. Be the software engineer who is replaced by AI because (1) you refuse to learn how to work with it, or (2) you blindly trust it and produce low-quality work.

There is no turning back; we accept it or perish.

- Humans have experienced similar technological revolutions before (e.g., the Industrial Revolution, the internet, computers, smartphones).
- Whatever our choices might be, these revolutions changed the way we work and live forever.

Software Engineering (Problem Solving) and AI

- Software engineering is about problem solving by managing complexity.
- Good software engineers are good at managing complexity using rules and tools.

Observations (and Questions)

- Can AI help bad software engineers become good software engineers by helping them manage complexity better?

What are your thoughts?

My Answer: Yes

- If the software engineers are bad because they lack knowledge or skills that AI can provide.
- In this case, AI is the miracle tool that can help them become good and 10x software engineers.

My Answer: No

- No: if the software engineers are bad because they lack critical thinking skills, creativity, or the ability to understand problems deeply.
- In other words, if they cannot define problems, make plans to solve them, and follow the plan to get results, AI (or any tool) cannot help them.

So, the Goal of this course is Clear

1. Students should learn to work with AI as a **dual brain** to enhance their problem-solving skills.
2. Students should learn to **remain in control** of their work, ensuring that they validate and oversee AI outputs.

All by **doing real software engineering projects** using AI as a cognitive partner!

All by **building real world IoT systems** using AI as a cognitive partner!

In Doing So

- Students will build the 2nd brain system that helps them think better, learn better, and work better.
- Students will learn how they can learn better (or worse) by working with AI as a cognitive partner.

The Role of the professor 1

- I will share my experience of working with AI as a cognitive partner.
 - I will share my tools & rules for working with AI effectively.
 - You will find your own tools & rules for working with AI effectively, and share them with your peers.

The Role of the professor 2

- I will show you IoT technologies and tools that you can use to build your projects.
 - You will make the transition from unknown unknowns to known unknowns.
- I will show you why the IoT technologies are good solutions to certain problem domains.
 - You will learn how to choose the right tools for your problems.

The Role of the student 1

- You define the problem you want to solve with "Why".
 - This course is not for you, if you can't define the problem you want to solve.
- You identify the technologies and tools you need to solve the problem.
 - Ask anything to AI (and me of course), for the answers.

The Role of the student 2

- You build the IoT system to solve the problem you defined.
 - You use AI as your cognitive partner to help you think better, learn better, and work better.
- You share your experience of working with AI as a cognitive partner with your peers.
 - You learn from your peers' experience of working with AI as a cognitive partner.

Project Evaluation

- Each student will be assigned three other students.
 - Each student will check these students' work and their progress.
 - The student should not know who are assigned to check his/her work.
- In the end, each student will evaluate the three students' work and their progress.

Next Steps

1. Make ready for the PPP (Project Proposal Presentation) in week 4.
2. Finish HW1 to be on the same page.
3. Request a meeting with me if you have anything to discuss with me.

Research Opportunities

1. For those of you who are interested in exploring more about AI, software engineering, and IoT, we can talk.
2. Doing Research and publishing papers together is one of the most effective ways to shine in your career as a software engineer, at the age of AI.

IoT Hardware

1. I will give each student an IoT hardware kit.
2. Each student should return the kit at the end of the course in the same condition.
3. If a student does not return the kit, they will earn failed grade (F) in this course & pay the full price of the kit.
4. If a student drops the course, they should return the kit immediately.